

Neha Ramachandran

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EDUCATION

Vellore Institute of Technology

B Tech in Computer Science 2nd year completed

Chennai, India

2024-2028

- CGPA: 8.70/10.0
- Relevant Coursework: Data Structures, Algorithms, Machine Learning, Artificial Intelligence, Database Management Systems, Full Stack Development

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, SQL, HTML/CSS

Frameworks: React.js, Next.js, Node.js, Express, FastAPI, Flask, PyTorch, Tailwind CSS

Tools & Databases: Git/GitHub, Linux, Postman, MySQL, MongoDB

Cloud/DevOps: Vercel, Render, Docker

ONGOING RESEARCH

Medical Deepfake Detection

Student Researcher

Chennai, India

Aug 2026 – Present

- Working with **Dr. Uday Tupakula** (Associate Professor, University of England) and **Dr. Geetha S** (Assistant Dean, Computer Science) to build an end-to-end deepfake detection pipeline for medical imaging using self-trained GAN-generated synthetic samples and a ResNet-based detector with strict leakage-free train/validation/test separation.
- Achieved strong **real vs GAN** synthetic classification performance with confusion matrices (CSV/PNG) and classification reports.
- Delivered reproducible benchmarking across **10+** datasets with standardized metrics and automated evaluation artifacts.

PROJECTS

Autism Spectrum Disorder Screening | PyTorch, Scikit-learn, SMOTE, Pandas

Summer 2025

- Developed a **machine learning pipeline** for early ASD detection using behavioral and demographic screening data, incorporating preprocessing, feature engineering, and class balancing techniques.
- Implemented and compared models including Logistic Regression, SVM, Random Forest, and a **Deep Neural Network** (DNN) with dropout and batch normalization.
- Achieved **93% accuracy** and **93.36% mAP**, demonstrating strong performance for scalable, non-invasive clinical screening.

Plant Disease Recognition Model | PyTorch, CNN, OpenCV, Python

2025

- Developed a deep learning-based image classification system to detect and classify plant diseases from leaf images using **convolutional neural networks (CNNs)**.
- Implemented data preprocessing, augmentation, and class balancing techniques to improve model generalization across varying environmental conditions.
- Achieved high classification accuracy through hyperparameter tuning and model optimization, enabling reliable early-stage disease detection.

Full-Stack Campus Social Platform | React.js, JavaScript, Node.js, Express, MySQL

2025

- Developed a **full stack** social media platform tailored for college communities, enabling students to connect, share posts, and interact within campus-specific networks.
- Built a responsive frontend using **React.js** and implemented RESTful APIs with **Node.js** and **Express** for handling user authentication, posts, and interactions.
- Designed and managed relational databases using MySQL (Workbench), ensuring efficient data storage, retrieval, and user relationship mapping.
- Implemented core features such as user profiles, post creation, likes, comments, and secure login systems.